PHYSICS

Stage 6

Module 2: Dynamics

Forces

Teacher: Dean Dimarti

Exam Equivalent Time: 30 minutes *(based on allocation of 1.5 minutes per mark)*

Blackwattle Bay

Questions

1. PHYSICS, M2 EQ-Bank 10178

A block is being dragged along a flat surface with a horizontal force of 30 N being applied to it.

The mass of the block is 6 kilograms.

If the block is travelling at a constant velocity, determine the magnitude of the frictional forces acting on the block, giving reasons for your answer. *(2 marks)*

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2. PHYSICS, M2 EQ-Bank 11068

A student pushes a 5.0 kg box across a rough horizontal surface with a force of 30 N. The box moves with constant velocity.

a. Identify the net force acting on the box. Justify your answer. *(1 mark)*

b. According to Newton's Third Law, describe the force pair involved in this situation. *(2 marks)*

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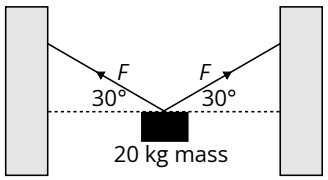
3. PHYSICS, M2 EQ-Bank 10801

The diagram below shows a 20 kg mass suspended from two strings.



Using a force diagram, calculate the force F in each string acting on the mass. *(3 marks)*

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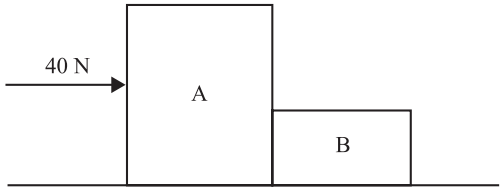
4. PHYSICS, M2 2018 VCE 8

Two blocks, A of mass 4.0 kg and B of mass 1.0 kg, are being pushed to the right on a smooth, frictionless surface by a 40 N force, as shown in the diagram.



a. Calculate the magnitude of the force on block B by block A $(F_{\text{on B by A}})$. Show your working. *(2 marks)*

b. State the magnitude and the direction of the force on block A by block B $(F_{\text{on A by B}})$. *(2 marks)*

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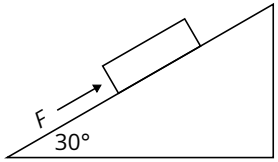
5. PHYSICS, M2 EQ-Bank 10356

A 2 kg crate is being pushed up a rough incline that is angled at 30° to the horizontal. The frictional force acting between the crate and the surface is 10 N.

a. On the diagram provided, draw and label all the forces acting on the crate as it is pushed up the incline. *(3 marks)*



b. Calculate the magnitude of the applied force F required to accelerate the crate up the incline at 0.8 ms^{-2} *(2 marks)*

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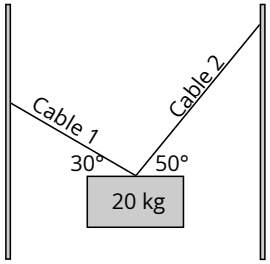
6. PHYSICS, M2 EQ-Bank 10890

A 20 kg crate is suspended in equilibrium by two cables as seen in the diagram below:



Determine the tension force (T_2) in Cable 2. *(3 marks)*

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Worked Solutions

1. PHYSICS, M2 EQ-Bank 10178

- As the block is travelling at a constant velocity, the sum of all the forces acting on the block must add to 0.
- The frictional forces must be equal in magnitude and opposite to the applied force.
- Therefore, the magnitude of the frictional forces is 30 N.

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2. PHYSICS, M2 EQ-Bank 11068

***Keyword (Part b.) - "Describe"**: Provide characteristics and features.*

a. The box moves at a constant velocity $\Rightarrow$ Net force acting on the box is zero.

b. The force pair is:

- The student exerts at force of 30 N on the box.
- The box exerts a reaction force of 30 N back on the student.

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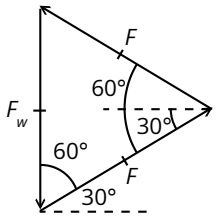
3. PHYSICS, M2 EQ-Bank 10801



- The weight force of the mass $F_w = mg = 20 \times 9.8 = 196\ \mathrm{N}$.
- As the force diagram is an equilateral triangle, the side length of all the sides of the triangle are equal.
- Hence the magnitude of $F_w = F = 196\ \mathrm{N}$.
- The force in each string is $196\ \mathrm{N}$.

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4. PHYSICS, M2 2018 VCE 8

a. Using $F = ma$, calculate the acceleration of the entire system:

Mean mark (a) 36%.

$$a = \frac{F}{m} = \frac{40}{5} = 8\ \text{ms}^{-2}$$

$$F_{\text{on B by A}} = m \times a = 1 \times 8 = 8\ \text{N}$$

b. Newton's third law of motion:

- $F_{\text{on A by B}}$ will be equal in magnitude and opposite in direction.
- $F_{\text{on A by B}} = 8\ \text{N}$ to the left.

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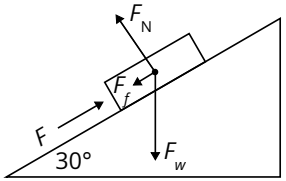
5. PHYSICS, M2 EQ-Bank 10356

a.



- The frictional force acts down the slope in this scenario because frictional forces oppose the direction of motion.

b. Weight force acting down slope $= mg\sin\theta = 2 \times 9.8 \times \sin 30 = 9.8\ \text{N}$

- Total force down the slope $= 9.8 + 10 = 19.8\ \text{N}$ (weight force down slope + frictional force)

$$F - 19.8 = ma_{\text{up slope}}$$

$$F = 2 \times 0.8 + 19.8$$

$$= 21.4\ \text{N}$$

$\therefore$ The force, (F), acting up slope needs to be $21.4\ \text{N}$.

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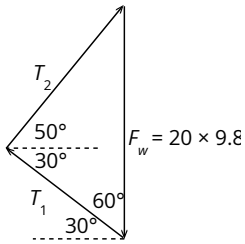
6. PHYSICS, M2 EQ-Bank 10890

- The weight force of the crate, $F_w = 20 \times 9.8 = 19.6\ \text{N}$ and so the following vector diagram can be set up below.



- Using the $\sin$ rule for finding the side length of a triangle, $\frac{a}{\sin A} = \frac{b}{\sin B}$

$$\frac{T_2}{\sin 60} = \frac{F_w}{\sin 80}$$

$$T_2 = \frac{19.6}{\sin 80} \times \sin 60$$

$$= 17.2\ \text{N}$$

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